# State Estimation and Filtering for Robotic Perception

**AA273**

Particle Filter



Stanford University

MSL

# AA273: State Estimation and Filtering for Robotic Perception

**Announcements:**

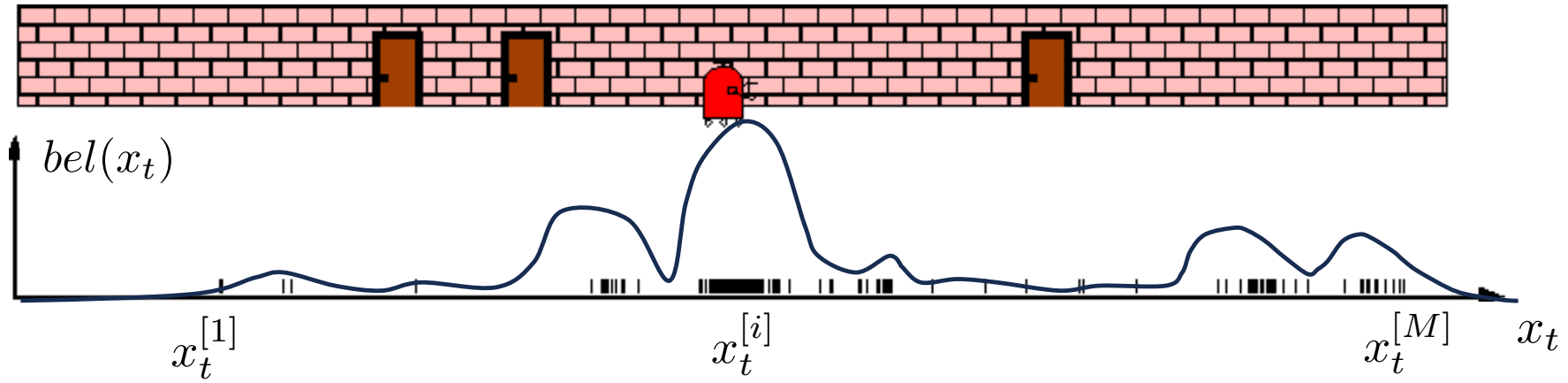
- HW 4 due Fri Feb 13
- HW 5 out Fri Feb 13, due Fri Feb 20
- Group signup sheet posted (groups from 1 – 4 people)
- Project Proposal: due Tue Feb 17
  - 2-3 pages, Latex
  - Related work >= 10 refs, ~1 page
  - Proposed work ~ 1 page
  - Bibtex refs

**Lecture 12:**

- Particle filter
- Importance Resampling
- Failure modes and mitigations

# Particle filter

- Key idea: represent posterior $bel(x_t)$ by a set of random samples
- Can represent arbitrary distributions
- Requires much more computation than KF/EKF




- Allows one to represent non-Gaussian distributions and handle nonlinear transformations in a direct way

# Particle filter

- Samples of posterior distribution are called *particles*, denoted as

$$\mathcal{X}_t = \{x_t^1, x_t^2, \ldots, x_t^N\}$$

- A particle represents a hypothesis about what the true world state might be at time *t*
- Ideally, particles should be distributed according to

$$x_t^i \sim p(x_t \mid y_{1:t}) = bel(x_t)$$

- Matching exact only as $N \to \infty$, but $N \approx 10^N - 100^N$ usually good enough for N dimensional state space
- A particle filter constructs the particle set $\mathcal{X}_t$ from the particle set $\mathcal{X}_{t-1}$ recursively, with the goal of matching the distribution $bel(x_t)$

# Recall: Nonlinear SS Model and Particle Filter

Dynamics eqn.:

$$X_t = f(X_{t-1}, u_{t-1}) + W_{t-1}$$

Or sample from equivalent transition dist.:

$$x_t^i \sim p(x_t \mid x_{t-1}^i, u_{t-1})$$

Measurement eqn.:

$$Y_t = g(X_t) + V_t$$

Or likelihood from equivalent meas. dist.:

$$w_t^i = p(y_t \mid x_t^i)$$

Independent white noise processes:

$$W_t \sim \mathcal{D}_W \qquad V_t \sim \mathcal{D}_V \qquad W_t, V_\tau \text{ independent for all } t, \tau$$

Given: $\mathcal{X}_0 = \{x_0^1, x_0^2, \ldots, x_0^N\}$ $\quad p(x_t \mid x_{t-1}, u_{t-1})$ $\quad p(y_t \mid x_t)$

For: $t = 1, 2, \ldots$

For: $i = 1, \ldots, N$

Prediction:

$$x_t^i \sim p(x_t \mid x_{t-1}^i, u_{t-1})$$

Update:

$$w_t^i = p(y_t \mid x_t^i)$$

Loop

Resample: For: $i = 1, \ldots, N$

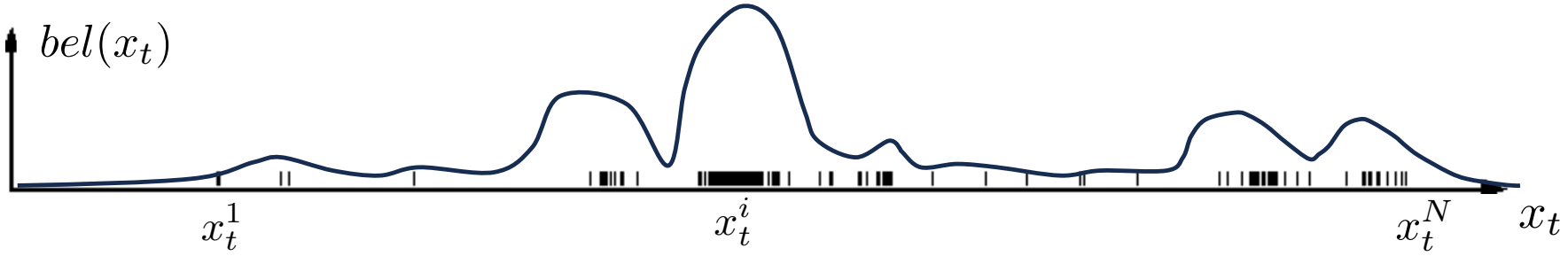
Draw $x_t^j$ with probability $\propto w_t^j$

Add $\overline{x}_t^i = x_t^j$ to new particle set $\mathcal{X}_t$

Loop

Loop

# Representing distributions as particle sets




The particles let us **approximate** any desired probability or moment:

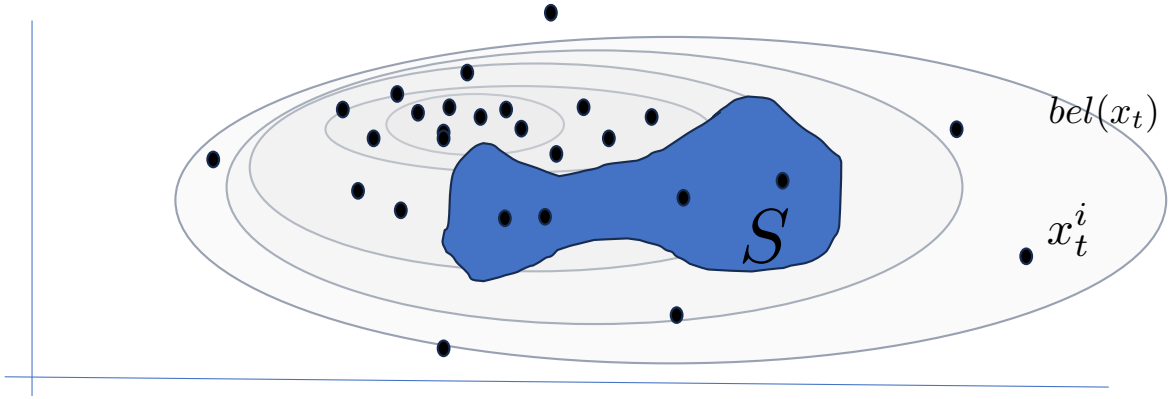
Mean: $$\mu_t \approx \frac{1}{N}\sum_{i=1}^{N} x_t^i = \tilde{\mu}_t$$

Covariance: $$\Sigma_t \approx \frac{1}{N}\sum_{i=1}^{N}(x_t^i - \tilde{\mu}_t)(x_t^i - \tilde{\mu}_t)^T = \tilde{\Sigma}_t$$

Probability:

$$Pr(x_t \in S) \approx \frac{\#\text{ particles in } S}{N} = \frac{|\{i \mid x_t^i \in S\}|}{N}$$

# Representing distributions as particle sets




The particles let us **approximate** any desired probability or moment:

Mean: $$\mu_t \approx \frac{1}{N}\sum_{i=1}^{N} x_t^i = \tilde{\mu}_t$$

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# Empirical vs Integral Expectations

Integral Expectation:

$$\mu_t = E[X_t] = \int_{x_t} x_t p(x_t)\, dx_t$$

$$\Sigma_t = Cov(X_t) = \int_{x_t} (x_t - \mu_t)(x_t - \mu_t)^T p(x_t)\, dx_t$$

$$Pr(x_t \in S) = \int_{x_t \in S} p(x_t)\, dx_t = \int_{x_t} \mathbf{1}_S(x_t) p(x_t)\, dx_t = E[1_S(x_t)]$$

$$E[f(X_t)] = \int_{x_t} f(x_t) p(x_t)\, dx_t$$

Empirical (a.k.a Monte Carlo) estimate:

$$\mu_t \approx \frac{1}{N} \sum_{i=1}^{N} x_t^i = \tilde{\mu}_t$$

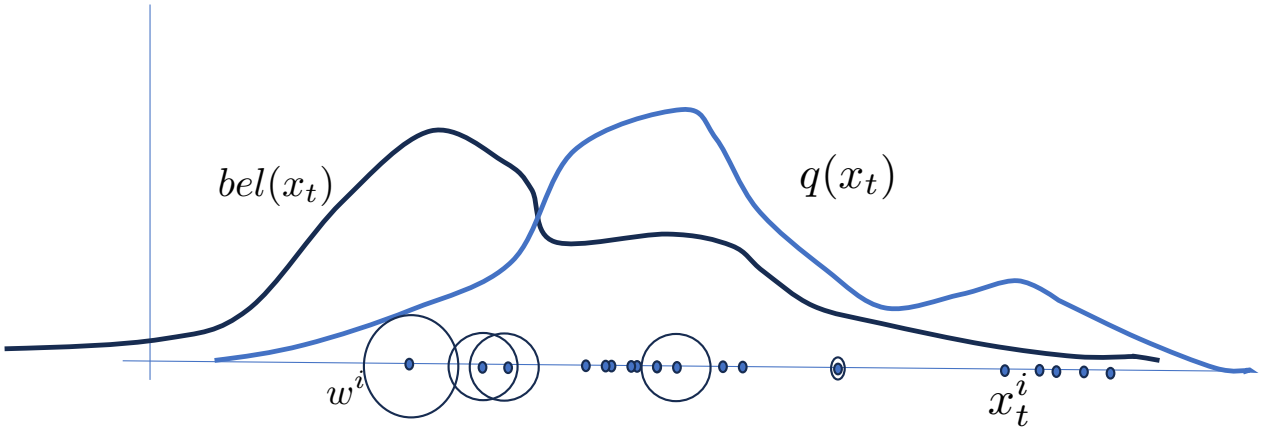
$$\Sigma_t \approx \frac{1}{N} \sum_{i=1}^{N} (x_t^i - \tilde{\mu}_t)(x_t^i - \tilde{\mu}_t)^T = \tilde{\Sigma}_t$$

$$Pr(x_t \in S) \approx \frac{\#\text{ particles in } S}{N} = \frac{|\{i \mid x_t^i \in S\}|}{N} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_S(x_t^i)$$

$$E[f(X)] \approx \frac{1}{N} \sum_{i=1}^{N} f(x_t^i)$$

**Law of Large Numbers:** $\lim_{N \to \infty} \left( \frac{1}{N} \sum_{i=1}^{N} f(X_i) \right) = E[f(X)]$ almost surely

# Importance Sampling




We want to represent $bel(x_t)$ with particles $x_t^{[m]} \sim q(x_t)$ from a different distribution

Weight the particles!

$$E_{bel}[f(x_t)] = \int_{x_t} f(x_t) bel(x_t)\, dx_t = \int_{x_t} f(x_t) \frac{bel(x_t)}{q(x_t)} q(x_t)\, dx_t$$

We have:

$$w(x) = \frac{bel(x)}{q(x)}$$

$$E_{bel}[f(x_t)] = \int_{x_t} f(x_t) w(x_t) q(x_t)\, dx_t \approx \sum_{x_t^i} f(x_t^i) w_t^i$$

Importance weights:

$$w_t^i = \frac{1}{N} \frac{bel(x_t^i)}{q(x_t^i)}$$

# Importance Sampling for Bayesian Filter

Claim, we can get weights without knowing $q(x_t)$ or $bel(x_t)$

On the board

# Discriminative Measurement Model

Measurement distribution:

$$p(y_t \mid x_t)$$

We **evaluate probabilities** from this model to get weights.

$$w_t^i = p(y_t \mid x_t^i)$$

Nonlinear state space form:

$$Y_t = g(X_t) + V_t$$

$V_t \sim \mathcal{D}_V(v_t)$ Non-Gaussian or Gaussian measurement noise

Plug $y_t, x_t^i$ into formula for $\mathcal{D}_V(y_t - g(x_t^i))$ to obtain $w_t^i$

# Discriminative Measurement Model

E.g., Gaussian measurement noise

$$V_t \sim \mathcal{N}(0, R_t) = \frac{1}{\sqrt{(2\pi)^n |R_t|}} e^{-\frac{1}{2} v_t^T R_t^{-1} v_t}$$

Plug $v_t = y_t - g(x_t^i)$ into

$$w_t^i = \frac{1}{\sqrt{(2\pi)^m |R_t|}} e^{-\frac{1}{2} (y_t - g(x_t^i))^T R_t^{-1} (y_t - g(x_t^i))}$$

To get $w_t^i$

# Generative Dynamics Model

State transition distribution:

$$p(x_t \mid x_{t-1}, u_{t-1})$$

We **draw samples** from this model to get new particles at time *t*.

$$x_t^i \sim p(x_t \mid x_{t-1}^i; u_{t-1})$$

Nonlinear state space model:

$$X_t = f(X_{t-1}, u_{t-1}) + W_{t-1}$$

$$W_{t-1} \sim \mathcal{D}_W(w_{t-1})$$ Non-Gaussian or Gaussian process noise

(1) Draw noise sample $w_{t-1}^i$ ← process noise, not particle weight! →

(2) Plug into nonlinear state space model to get $x_t^i = f(x_{t-1}^i, u_{t-1}) + w_{t-1}^i$

This draws $x_t^i$ from the state transition distribution $x_t^i \sim \mathcal{D}_W(x_t - f(x_{t-1}^i, u_{t-1}))$

# Generative Dynamics Model

E.g., Gaussian process noise

(1) Draw: $w_{t-1}^i \sim \mathcal{N}(0, Q_t)$

(2) Evaluate: $x_t^i = f(x_{t-1}^i, u_{t-1}) + w_{t-1}^i$

Equivalent to drawing from state transition distribution:

$$x_t^i \sim \frac{1}{\sqrt{(2\pi)^n |Q_t|}} e^{-\frac{1}{2}(x_t - f(x_{t-1}^i, u_{t-1}))^T Q_t^{-1} (x_t - f(x_{t-1}^i . u_{t-1}))}$$

# Recall: Nonlinear SS Model and Particle Filter

Dynamics eqn.:

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Or sample from equivalent transition dist.:

$$x_t^i \sim p(x_t \mid x_{t-1}^i, u_{t-1})$$

Measurement eqn.:

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Or likelihood from equivalent meas. dist.:

$$w_t^i = p(y_t \mid x_t^i)$$

Independent white noise processes:

$$W_t \sim \mathcal{D}_W \qquad V_t \sim \mathcal{D}_V \qquad W_t, V_\tau \text{ independent for all } t, \tau$$

Given: $\mathcal{X}_0 = \{x_0^1, x_0^2, \ldots, x_0^N\}$ $\quad p(x_t \mid x_{t-1}, u_{t-1})$ $\quad p(y_t \mid x_t)$

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Update:

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Loop

Resample: For: $i = 1, \ldots, N$

Draw $x_t^j$ with probability $\propto w_t^j$

Add $\overline{x}_t^i = x_t^j$ to new particle set $\mathcal{X}_t$

Loop

Loop

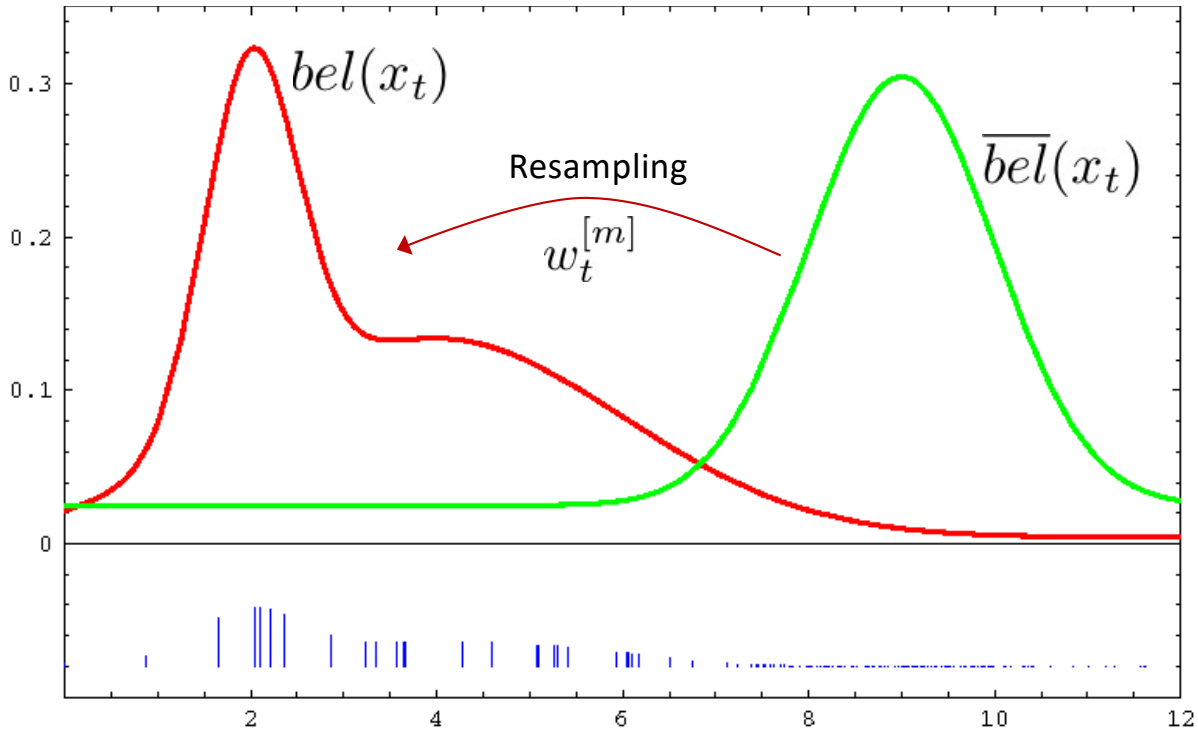
# Resampling

Draw particle i with probability $\propto \omega_t^{[i]}$

On the board

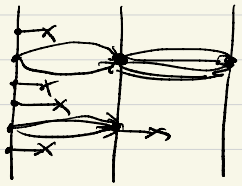
# Resampling

- Variants exist to sample less frequently (not every time step) to avoid particle diversity problems.



# Failure Modes and Remedies

(a) Particle Impoverishment




cause: resample too often

solution: resample less often

$\{x_{t|t-1}^i, w_t^i\}_{i=1}^N$

↳ no resample

$\{x_{t|t}^i, w_t^i\}_{i=1}^N$

$\|$

$x_{t|t-1}^i$ (keeps weighted particles)

at update:

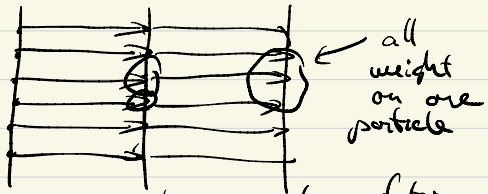
$\hat{w}_{t+1}^i = p(Y_{t+1} = y_{t+1} \mid X_{t+1} = x_{t+1|t}^i) \cdot w_t^i$

$w_{t+1}^i = \dfrac{\hat{w}_{t+1}^i}{\sum_{i=1}^N \hat{w}_{t+1}^i}$

proceed.

(b) Particle Degeneracy

opposite problem




cause: not resampling often enough.

tests for these failure modes:

(a) if $\frac{1}{N}\sum_{i=1}^N \| x_{t|t}^i - \mu_{t+1|t} \|^2 \leq$ thresh

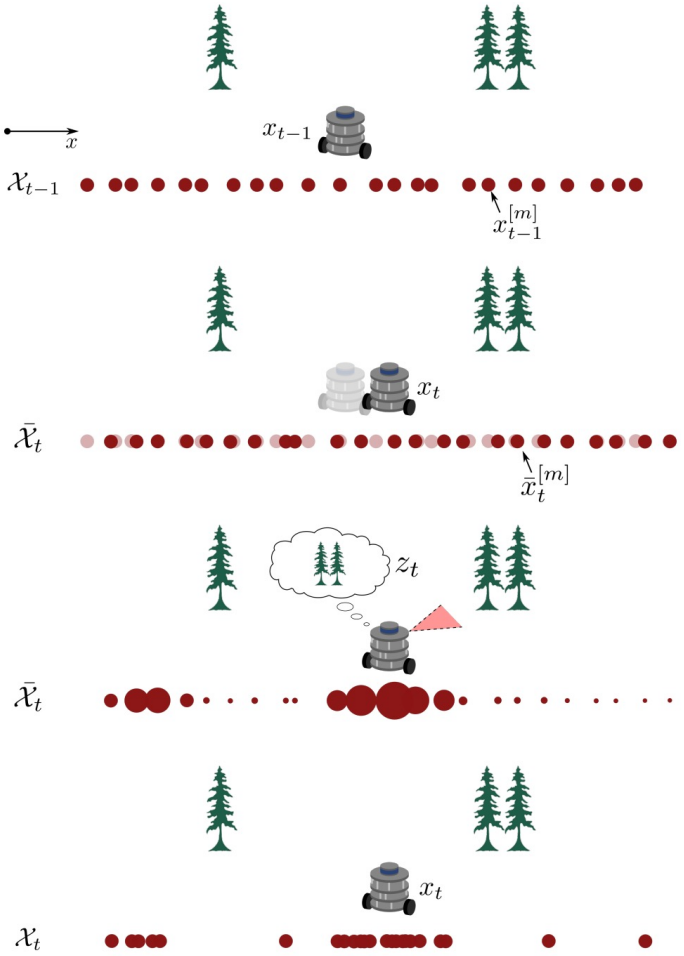
(unweighted average)

then don't resample

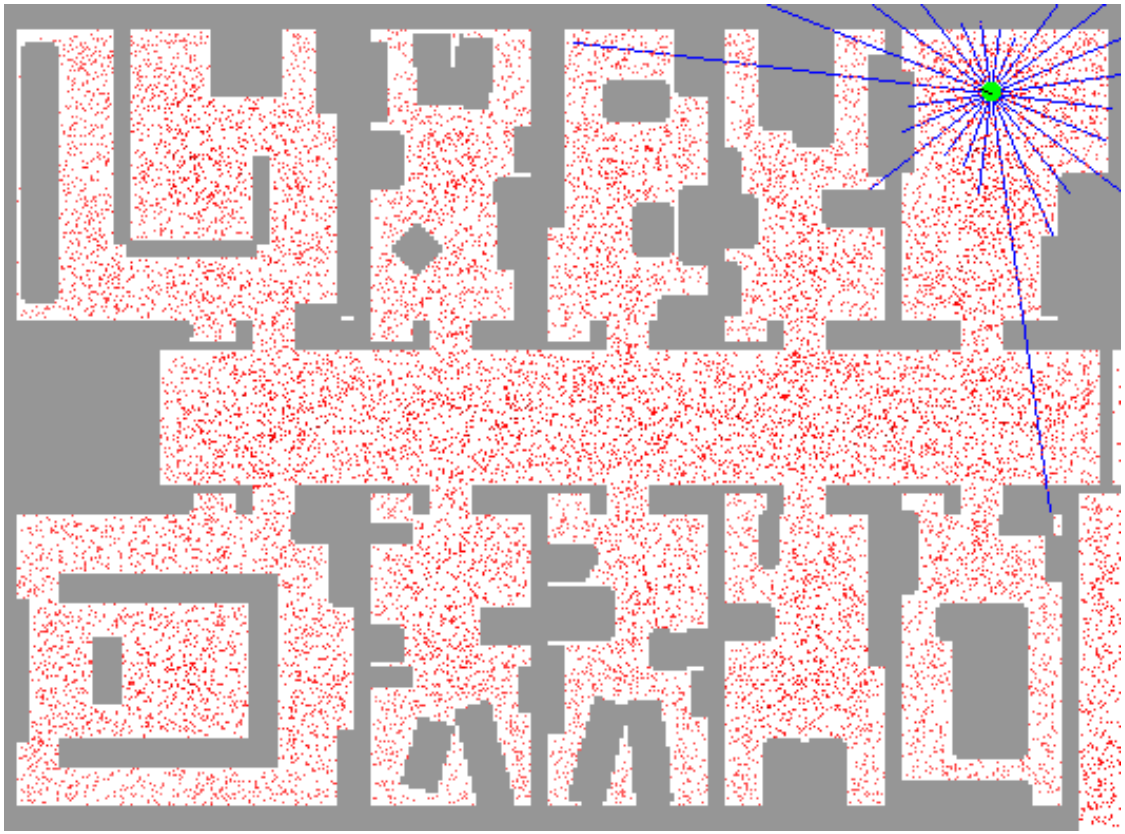
(b) if $\frac{1}{N}\sum_{i=1}^N (w_t^i - \frac{1}{N})^2 \geq$ thresh

then do resample

# Particle filter: example

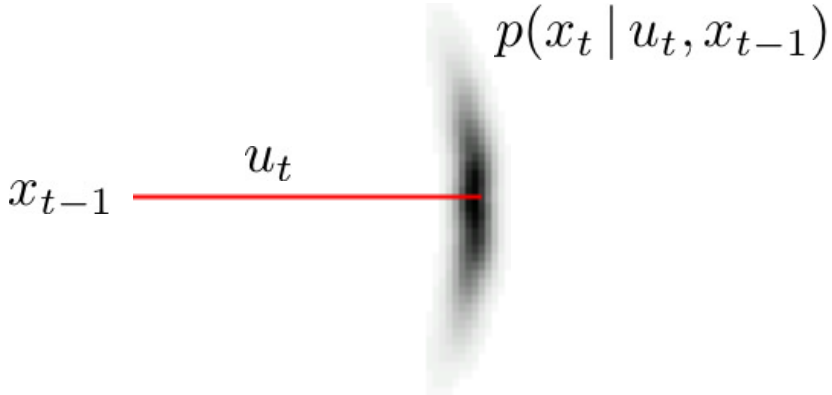


# MCL: example



# E.g.: Monte Carlo localization

- Motion model is probabilistic




PF requires us to sample:

$$x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$$

Example for dynamics with additive noise:

$$x_t^{[m]} = f(x_{t-1}^{[m]}, u_t) + \epsilon_{t-1}^{[m]}, \quad \epsilon_{t-1}^{[m]} \sim \mathcal{N}(0, R_{t-1})$$

Also requires to evaluate measurement likelihood:

$$w_t^{[m]} = p(z_t \mid x_t^{[m]})$$

E.g., for Gaussian:

$$w_t^{[m]} = \mathcal{N}(C_t x_t^{[m]}, Q_t)$$

# Monte Carlo localization

- Key idea: represent belief $bel(x_t)$ by a set of *M* particles

$$\mathcal{X}_t = \{x_t^{[1]}, x_t^{[2]}, \ldots, x_t^{[M]}\}$$

- Requires a map *m* as input
- Can handle dynamic environments via outlier rejection

Compute for occupancy map
Or feature based map

**Data:** $\mathcal{X}_{t-1}, u_t, z_t, m$
**Result:** $\mathcal{X}_t$
$\overline{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$;
**for** $i = 1$ **to** $M$ **do**
  Sample $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]}, m)$;
  $w_t^{[m]} = p(z_t \mid x_t^{[m]}, m)$;
  $\overline{\mathcal{X}}_t = \overline{\mathcal{X}}_t \cup \left(x_t^{[m]}, w_t^{[m]}\right)$;
**end**
**for** $i = 1$ **to** $M$ **do**
  Draw $i$ with probability $\propto w_t^{[i]}$;
  Add $x_t^{[i]}$ to $\mathcal{X}_t$;
**end**
Return $\mathcal{X}_t$

Sample reject collisions